Final presentation

Electricity Price Trends and Market Volatility in Europe

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

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Project

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I. Executive Summary
II. Preprocessing
III. Cluster Analysis
IV. Price Predictions
V. Recommended Next Steps

Executive Summary

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Initial Situation

Marbl faces increasing market volatility and needs deeper insights into recurring price patterns to improve data-driven trading decisions and forecasting accuracy

Project Scope

Generate **market-intelligence** insights by identifying and visualizing recurring day-ahead price-pattern clusters for **DK1**, **ES** and **NO2**

Build a cluster-based **forecasting** model with a visualization **dashboard** for day-ahead **predictions**

Methodology

- Automated data Integration
- Preprocessing
- Pattern Detection
- Predictions

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Preprocessing

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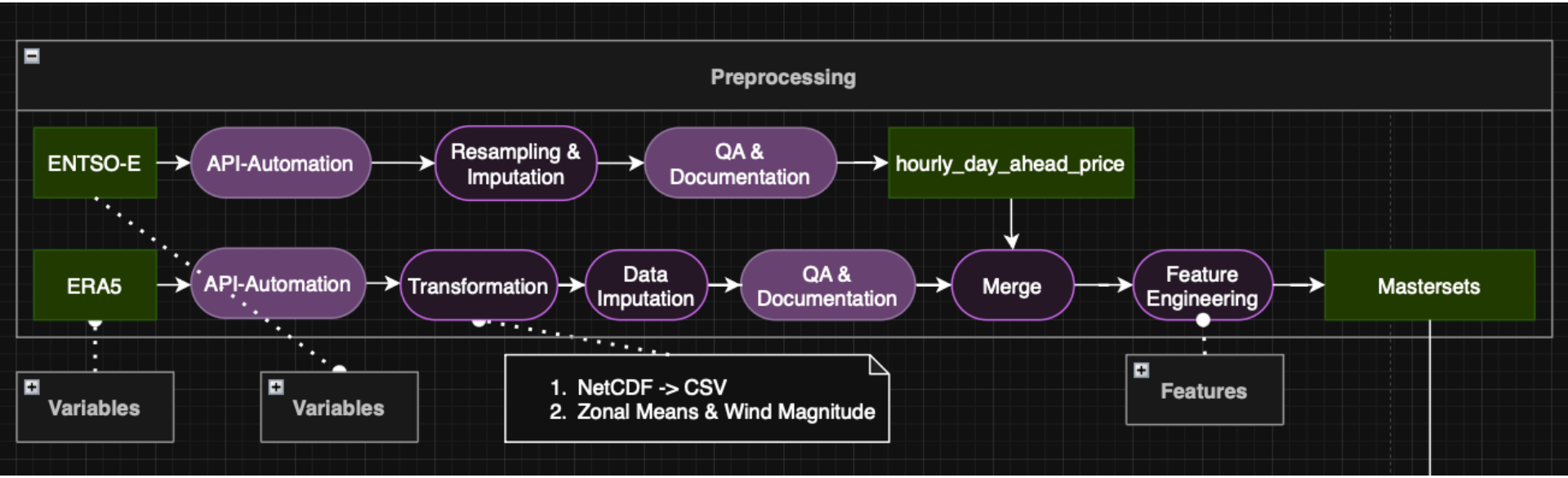
I. Structure

II. ENTSO-E

III. ERA5

Structure


Preprocessing
ENTSO-E
API-Automation
Resampling & Imputation
QA & Documentation
hourly_day_ahead_price
ERA5
API-Automation
Transformation
Data Imputation
QA & Documentation
Merge
Feature Engineering
Mastersets
Variables
Variables
1. NetCDF -> CSV
2. Zonal Means & Wind Magnitude
Features

ENTSO-E (Track A)

- European Transparency Platform: The central open-data hub for all European Transmission System Operations (TSOs)

- Data Scope:
  - Variable: Day-Ahead-Prices (EUR/MWh)
  - Zones:
    - Denmark West (DK1): Wind-dominated market
    - Spain (ES): Solar-heavy-market
    - Southern Norway (NO2): Hydro-storage-market
  - Timeline: 2015 - 2025

- Implementation
  - Access: Automated ingestion via entsoe-py library, ensuring standardised data retrieval

ENTSO-E – Challenges & Solutions

- API Complexity & Format
  - Challenge: Raw XML parsing was error-prone
  - Solution: Library Integration (EntosePandasClient) for standardised retrieval

- Query Limits & Stability
  - Challenge: 10-year requests cause timeouts
  - Solution: Smart Chunking (6-month iterative blocks)

- Granularity Mismatch
  - Challenge: most bidding zones data shifted from hourly to 15-min resolution
  - Solution: Auto-Resampling to hourly means

ERA-5 (Track B)

- Global atmospheric reanalysis dataset produced by ECMWF
  - Spatial resolution: ~31 km (0.25° × 0.25°)
  - ~ 5-day delayed update
  - Variables (hourly)
    - Temperature 2m above ground
    - Windspeed
    - Irradiance
    - Precipitation

- Dataformats
  - GRIB
  - NetCDF

ERA-5 – Challenges & Solutions

- API Constraints
  - Challenge: Low account priority & rate limits
  - Solution: Robust Retry Logic with automated verification

- Spatial Mismatch
  - Challenge: Square grids vs. irregular bidding zones
  - Solution: Zonal Spatial Aggregation (shapefile overlay -> zonal Mean)

- Feature Readiness
  - Challenge: Raw vectors (U/V) not model-ready
  - Solution: Physics-based Feature Engineering
    - Technique: Calculating Wind Speed Magnitude $(Speed = \sqrt{(u^2 + v^2)})$

Cluster Analysis

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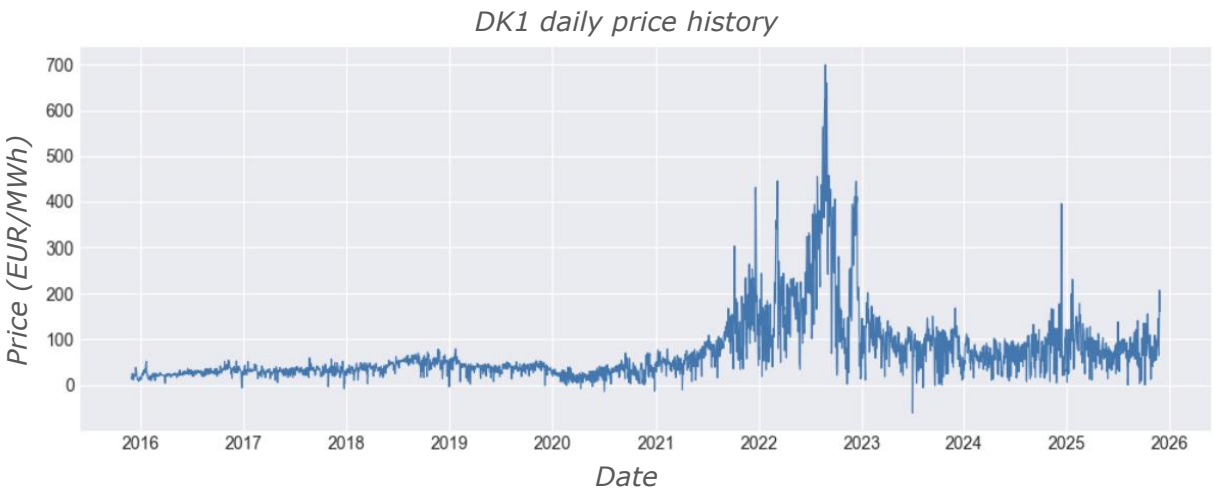
I. Exploratory Data Analysis
II. Structure
III. Deep Dive: DK1 Cluster

Exploratory Data Analysis



Same Pattern for **ES** & **NO2**

Long term **price histories** show clear structural **shifts**, and price levels and volatility **today differ** fundamentally from **ten years** ago

To **avoid mixing** outdated regimes with current dynamics, we **limit** the clustering to the most **recent three years** with consistent market behavior.

Cluster Architecture

Hourly day-ahead-prices
- Hourly electricity prices per bidding zone

↓

Daily profile vectors
- Each day represented as a 24-hour price profile

↓

Hierarchical clustering
- Ward linkage with Euclidean distance to group similar days

↓

Cluster cut selection
- Elbow plot, acceleration plot, and cluster size distribution

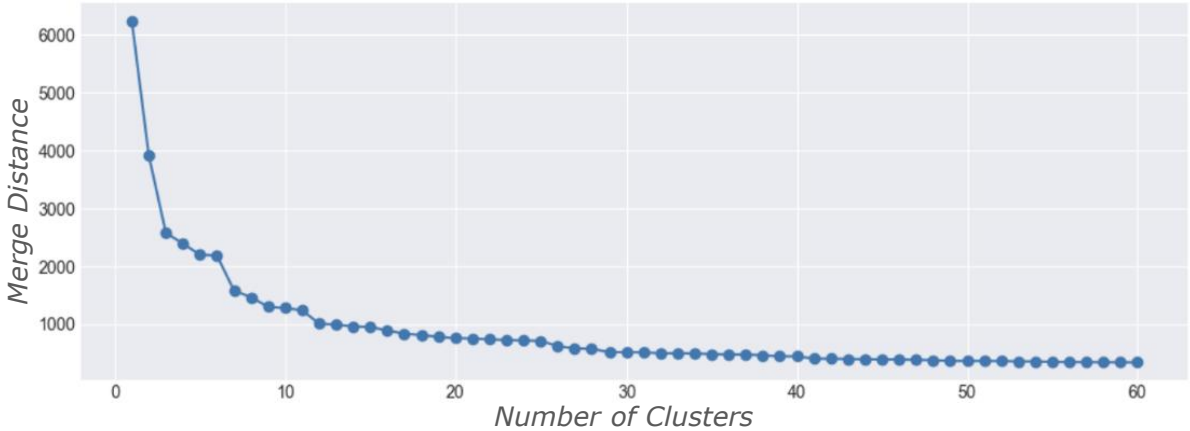
↓

Market regime outputs
- Cluster labels, centroid profiles, and calendar mapping

↓ **Market intelligence insights**

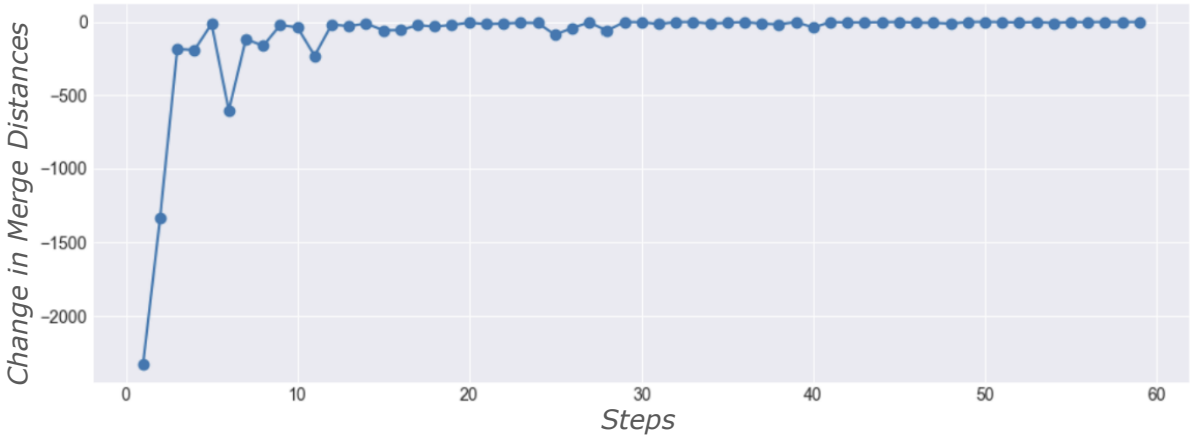
↓ **Input for prediction models**

Deep Dive: DK1 Cluster Selection 1/2



Elbow plot

Shows where **adding more** clusters **no** longer **leads** to a strong **reduction** in merge **distances**.



Acceleration plot

Shows where **large jumps** in merge **distances** occur **during** the **clustering** process.

Deep Dive: DK1 Cluster Selection 2/2

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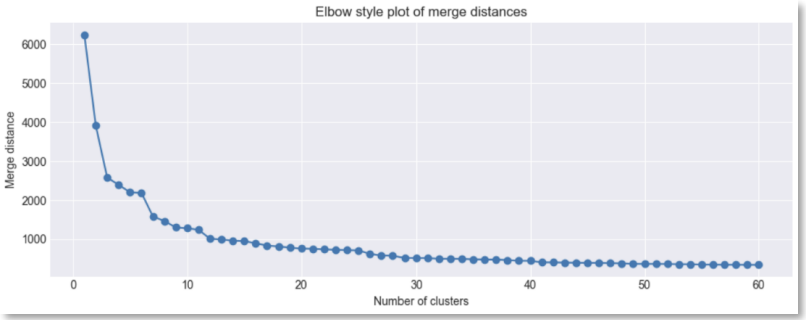
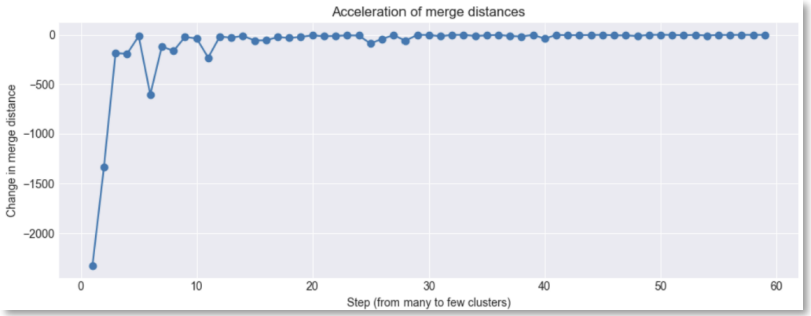
Elbow plot



Acceleration plot

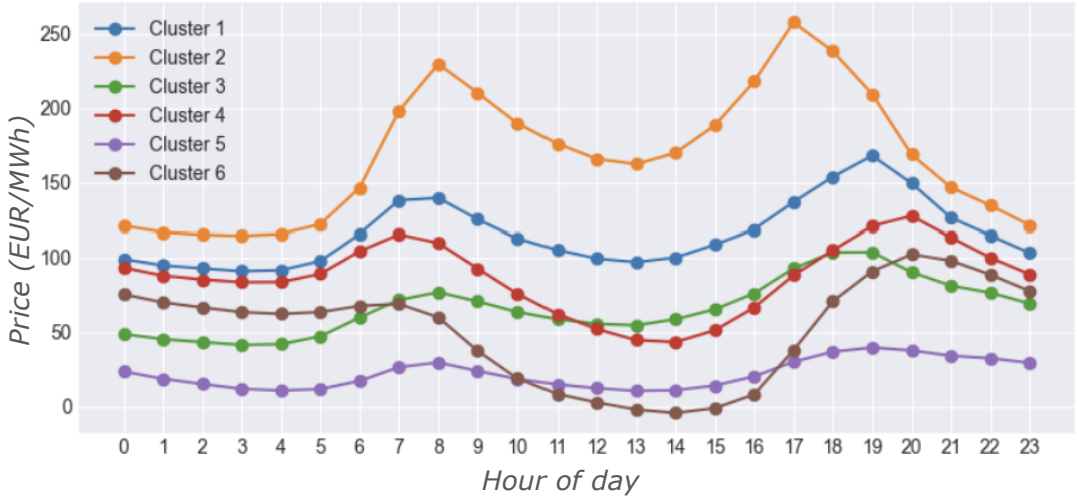


Cluster Distribution

Cluster	Number of days
Cluster 1	222
Cluster 2	45
Cluster 3	187
Cluster 4	308
Cluster 5	119
Cluster 6	180

The final **cluster count** is chosen where the **plots** indicate clear **structural changes** and the cluster **distribution** remains **stable**.

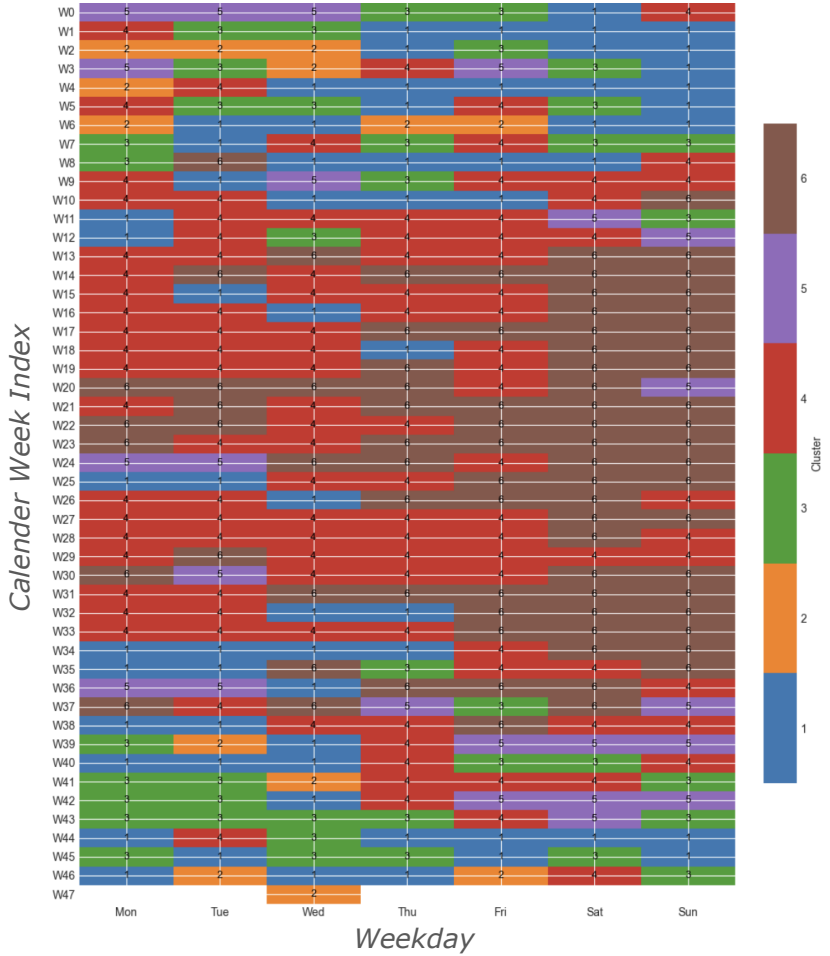
Deep Dive: DK1 Visualisation 1/2



Centroid chart

shows the **average hourly** price **profile** of each **cluster**, representing a typical **intraday** price **pattern**

Deep Dive: DK1 Visualisation 2/2

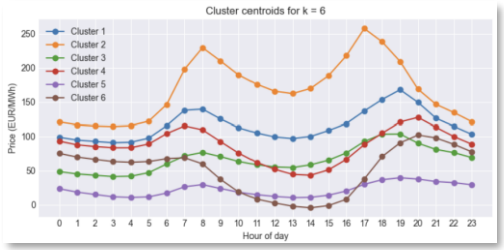


Cluster calendar

shows **when** each price **regime occurs** by **assigning** every **day** of the year to its corresponding **cluster**.

By displaying the calendar over **three years**, **recurring** seasonal and weekly **patterns** become visually **identifiable**.

Deep Dive: DK1 Interpretation



Cluster	Number of days
Cluster 1	222
Cluster 2	45
Cluster 3	187
Cluster 4	308
Cluster 5	119
Cluster 6	180

Interpretation

Summer shows long stretches of the red cluster

Spring and autumn show the widest cluster mix

Weekends often fall into the brown volatile cluster

Orange is rare and marks short high-volatility periods

Recent market **volatility blurs** traditional price **regimes** and **limits** the **interpretability** of statistically robust clusters.

Price Predictions

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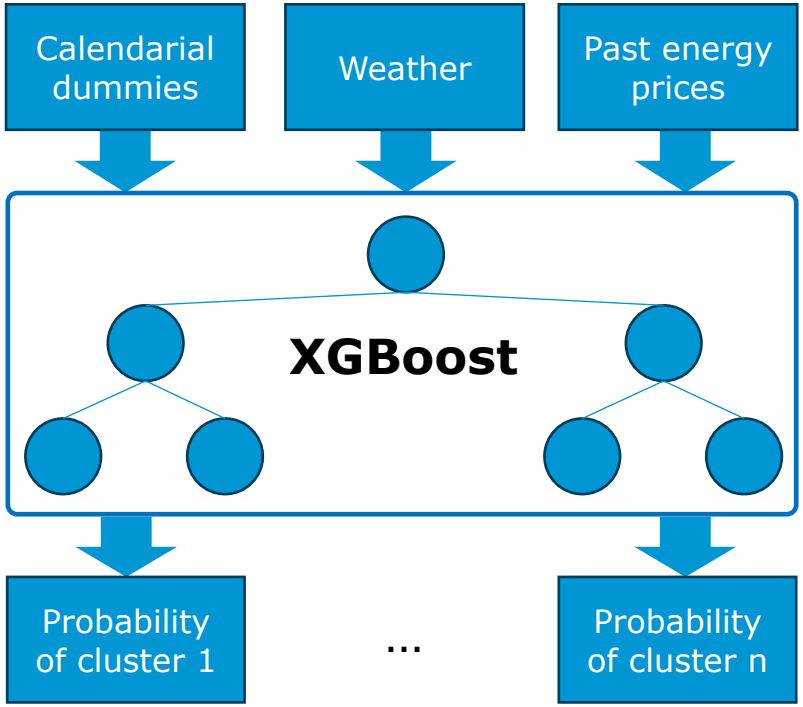
I. Prediction Modelling
   I. Cluster Predictions
   II. Hourly within Cluster Predictions
   III. Final Price Prediction
II. Results

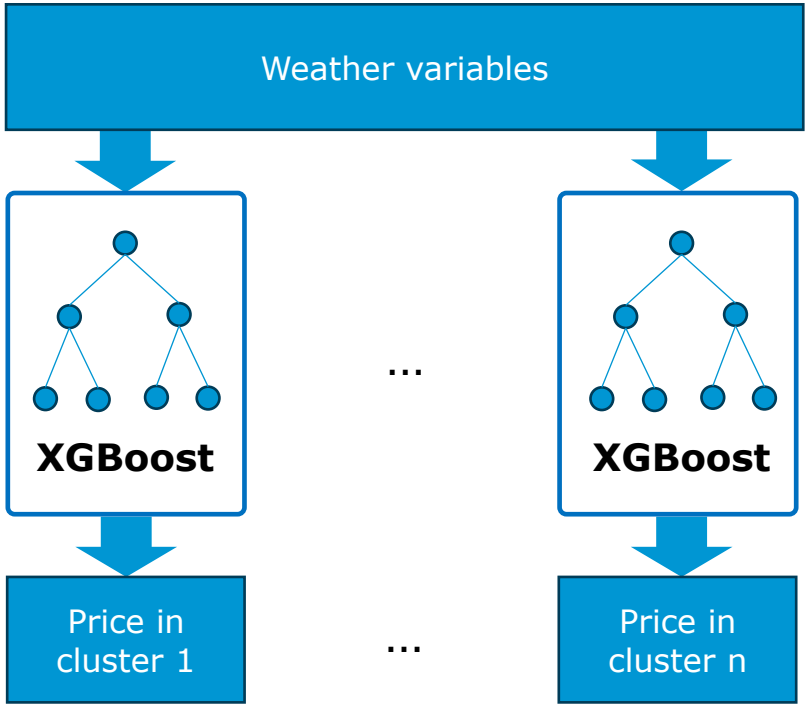
Cluster Predictions

- **Goal**: Predict the probabilities of the different possible clusters for a day
- **Input**: Calendarial dummies (weekend/weekday, season) and weather variables (mean temperature, mean wind speed, precipitation, irradiance) and past energy price levels (last 3 days)
- **Model**: XGBoost classification
- **Output**: Probabilities of occurence for the different clusters identified within a bidding zone for a day



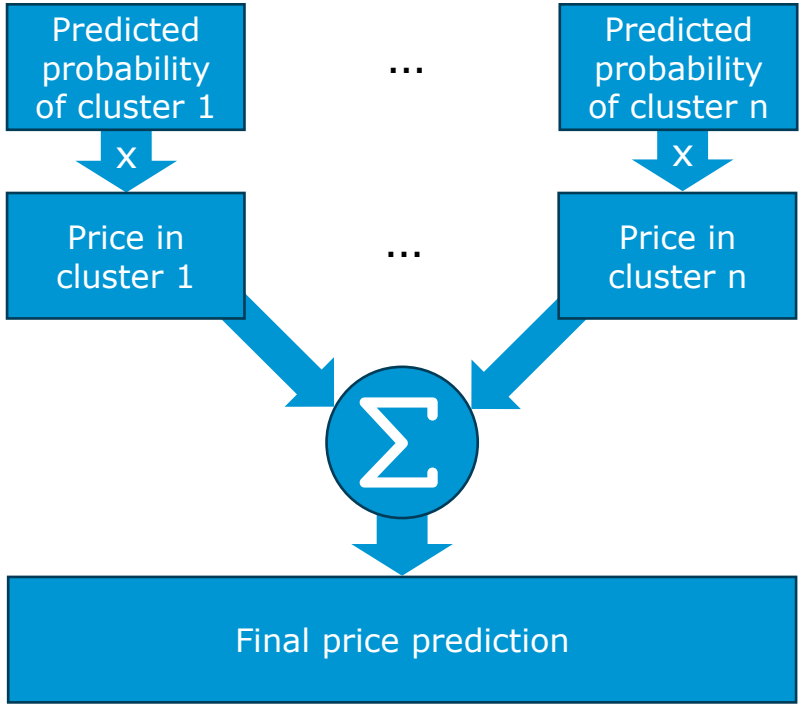
Hourly within Cluster Predictions

- **Goal**: Predict the hourly whole sale energy prices for each cluster
- **Input**: Hourly values for 3/4 weather variables (temperature, wind speed, irradiance) and aggregated precipitation from the last x days (x ∈ {3, 7, 14, 20})
- **Model**: XGBoost regression (one model for each cluster from the cluster analysis and bidding zone ($\sum = 56$))
- **Output**:
  - One value per hour and cluster
  - E.g. 6 clusters in DK1 lead to 6 energy prices



Final Price Prediction

- **Goal**: Obtain one final prediction for the energy price per hour
- **Input**: Predicted probability of each cluster and the predicted price for each cluster
- **Calculation:** Weigh the predicted prices of all clusters with the according predicted cluster probabilities and sum up the single values
- **Output**: One final energy price prediction per hour



Results

- To measure the performance of our cluster-based approach, we agreed on two naive approach benchmarks with Marbl.
- **Naive approach 1**: Utilize the day-before prices as predictions for the current day.
- **Naive approach 2**: Directly leverage the chosen weather variables as covariates, without the cluster layer.
- **Chosen metric**: Weighted Absolute Percentage Error (WAPE)

	Bidding zone	WAPE [%]
Cluster-based models	**DK1**	**31.45**
	ES	33.84
	NO2	23.28
Naive approach 1	DK1	34.39
	ES	**29.60**
	NO2	**22.46**
Naive approach 2	DK1	33.56
	ES	40.70
	NO2	30.49

Dashboard

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I. Demo

The MARBL Energy Dashboard

- https://dslab25wmarblenergy-elbldzbhtpnpwxgbjt3gax.streamlit.app

MARBL Dashboard | Downloadverlauf

localhost:8501

Deploy

app
Market Overview
Cluster Analysis
Live Forecast

marbl.

Navigation

Use the pages above to explore the dashboard.

MARBL Energy Dashboard

Electricity Price Trends and Market Volatility in Europe

This dashboard provides insights into European electricity markets, including historical price analysis, pattern detection, and day-ahead price forecasting for three distinct bidding zones.

Bidding Zones

Denmark West (DK1)

Wind Market

Wind-dominated market with high renewable penetration (55% wind generation) | Data Available

Spain (ES)

Solar Market

Solar-dominated market with significant midday price dips from PV generation | Data Available

South Norway (NO2)

Hydro Market

Hydro-dominated market connected to Continental Europe via Skagerrak cables | Data Available

Dashboard Pages

Market Overview

Explore historical price and weather data. View price trends, daily profiles, and correlations between weather conditions and electricity prices.

Cluster Analysis

View daily load price patterns from hierarchical clustering. Understand recurring daily price behaviors and their seasonal distribution.

Live Forecast

Day-ahead price predictions based on weather forecasts and the XGBoost Mix-of-Experts model.

About This Project

This dashboard was developed as part of the **Data Science Lab 2025/26 at WU Vienna** in collaboration with **Marbl FlexCo**.

The project investigates electricity price dynamics across three European bidding zones with distinct generation mixes: Denmark (wind-dominated), Spain (solar-dominated), and Norway (hydro-dominated).

Key Components:

- Historical price analysis and weather correlation
- Hierarchical clustering for pattern detection
- XGBoost Mix-of-Experts forecasting model

> Project Team

Data Sources: Electricity prices from ENTSO-E Transparency Platform | Historical weather from ERA5 Reanalysis (Copernicus Climate Data Store) | Weather forecasts from WeatherAPI.com

Disclaimer: This dashboard is for educational and research purposes only. Forecasts are model predictions and should not be used for trading decisions.

Recommendations & Future Work

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I. Recommendations & Future Work

Recommended for Follow-Up Projects

- Performance Analysis
  - Naive vs. ML: Investigate specific regimes where Naive models outperform the cluster-based approach
  - DK1 Success: Analyze the specific market drivers behind the superior performance in Denmark
- Model & Data Enhancement
  - New Features: Integrate Generation Capacity and Hydro-Reservoir levels
  - Benchmarks: Test Deep Learning architectures (e.g., LSTM, Transformers) against XGBoost
- System Expansion
  - Geography: Leverage modular pipeline to add Germany (DE-LU) and Netherlands (NL)
  - Dashboard Interactivity: Add "What-If" analysis to simulate price impacts from changing weather inputs

Appendix

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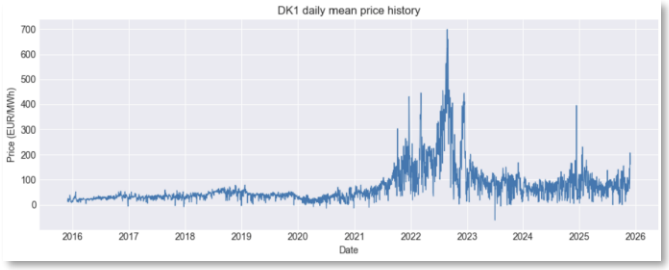
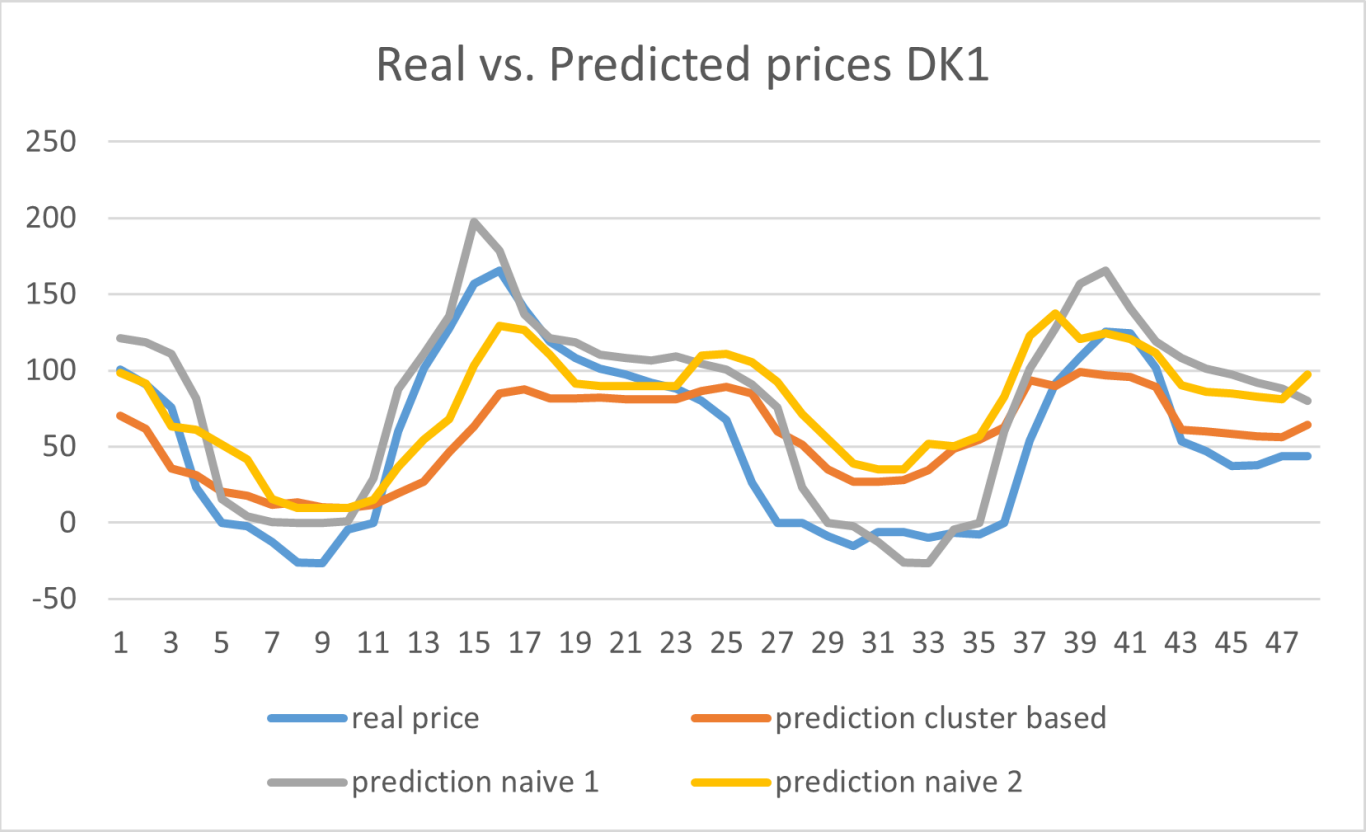
I. Full EDA

Full Exploratory Data Analysis



Real vs. Predicted Prices

- **Timespan**: 48 hours (June 21st – June 23rd 2025)
- **Bidding Zone**: DK1
- **Unit**: €/MWh



Weighted Absolute Percentage Error

- $WAPE = \frac{\sum_{t=1}^{n} |y_t - \hat{y}_t|}{\sum_{t=1}^{n} |y_t|} * 100\%$

- y_t … *real value at time t*
- $\hat{y}_t$ … *predicted value at time t*
- n … *number of observations*

- WAPE provides a robust, stable and volume-weighted measure for forecast quality, directly reflecting business impact for energy trading companies (Chockalingam, 2023)
- Scale free and easy to interprets
- Robust to volatility and outliers
- Avoids near-zero denominator problem of pointwise percentage errors